
GRAFIKA KOMPUTEROWA - LAB 13

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Joanna Dagil 231008

```
clear; close all; clc;
```

CEL CWICZENIA

Filtrowanie sygnałów jedno- i dwuwymiarowych.

ZADANIA

```
% -----
```

ZADANIE 1

```
-----  
[music,f] = audioread('icing2.wav');  
[K,~] = size(music);  
fmusic = fft(music);  
% zamiana wsp. x na herce  
for i = 1:K  
    hz(i) = f*(i-1)/K;  
end  
hz2 = hz-22050;  
fmusic2 = fftshift(fmusic);  
figure(1); plot(hz2,abs(fmusic2));  
% pierwszy filtr  
M = 100;  
OP = ones(1,M)/M;  
[h,w] = freqz(OP,1,512);  
figure(2); plot(f*w/(2*pi),abs(h));  
music_out = filter(OP,1,music);
```

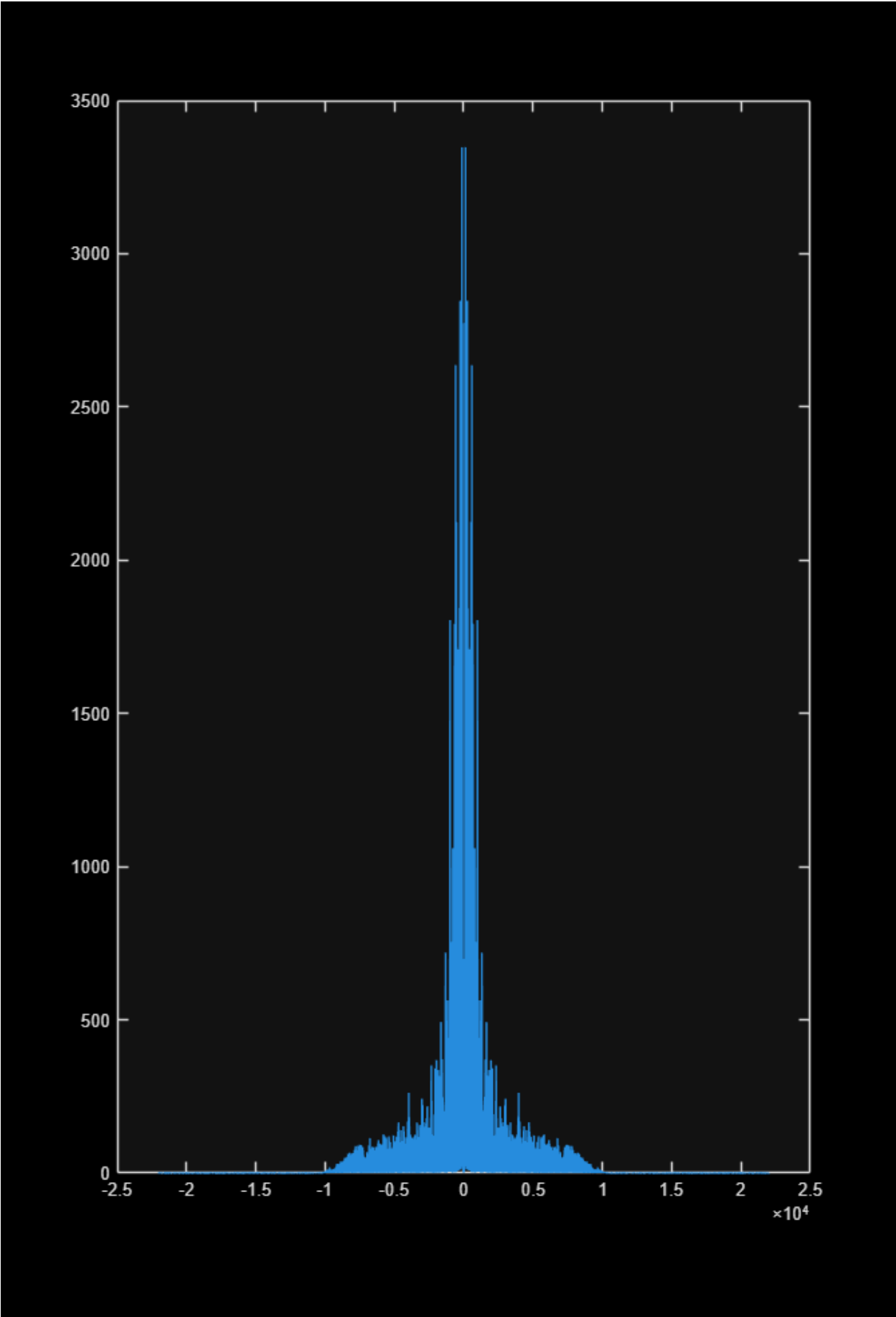
```
audiowrite('icing2_avg.wav', music_out, f);
```

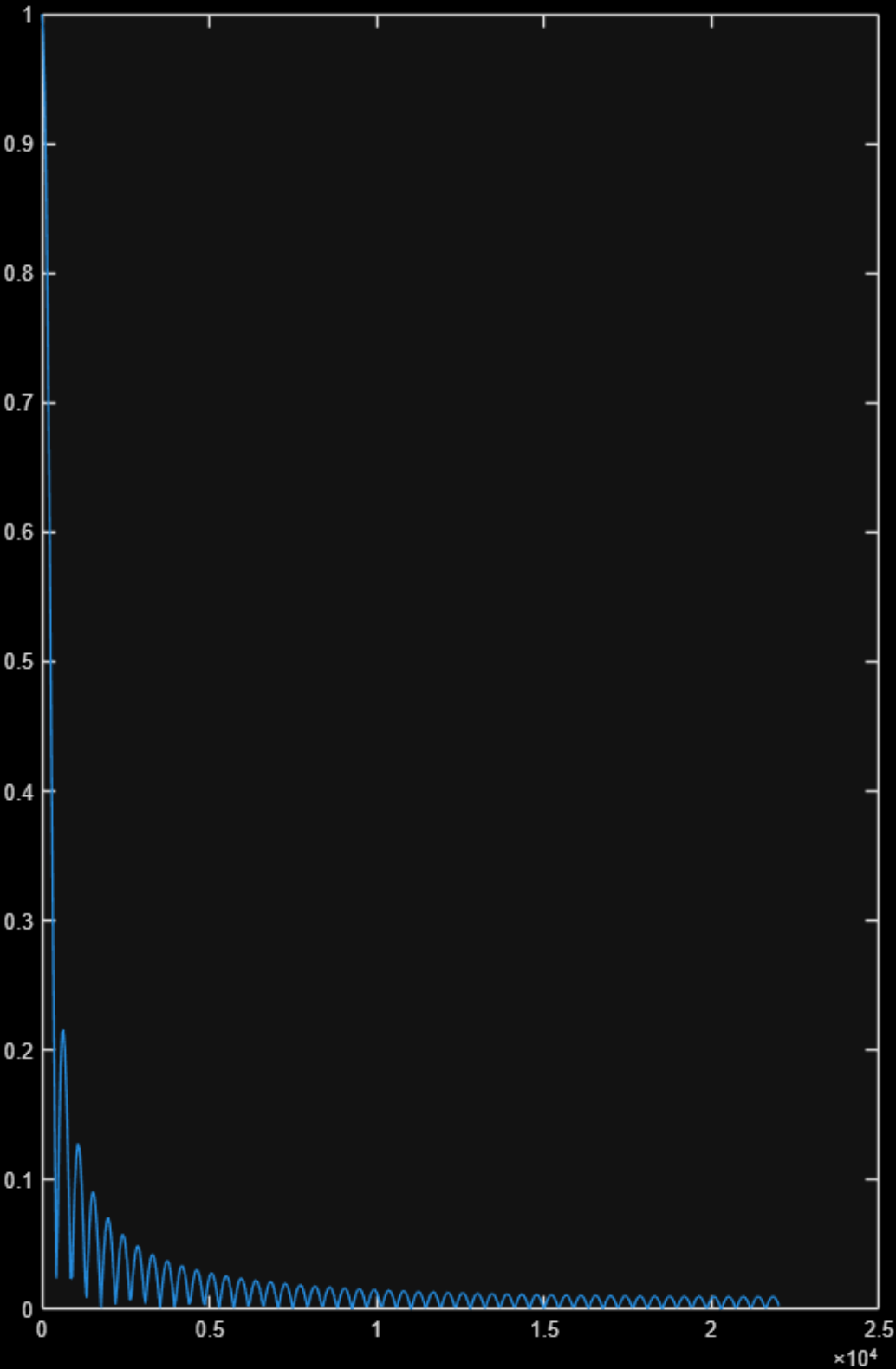
```
fmusic_out = fft(music_out);  
fmusic_out2 = fftshift(fmusic_out);  
figure(3); plot(hz2,abs(fmusic_out2));
```

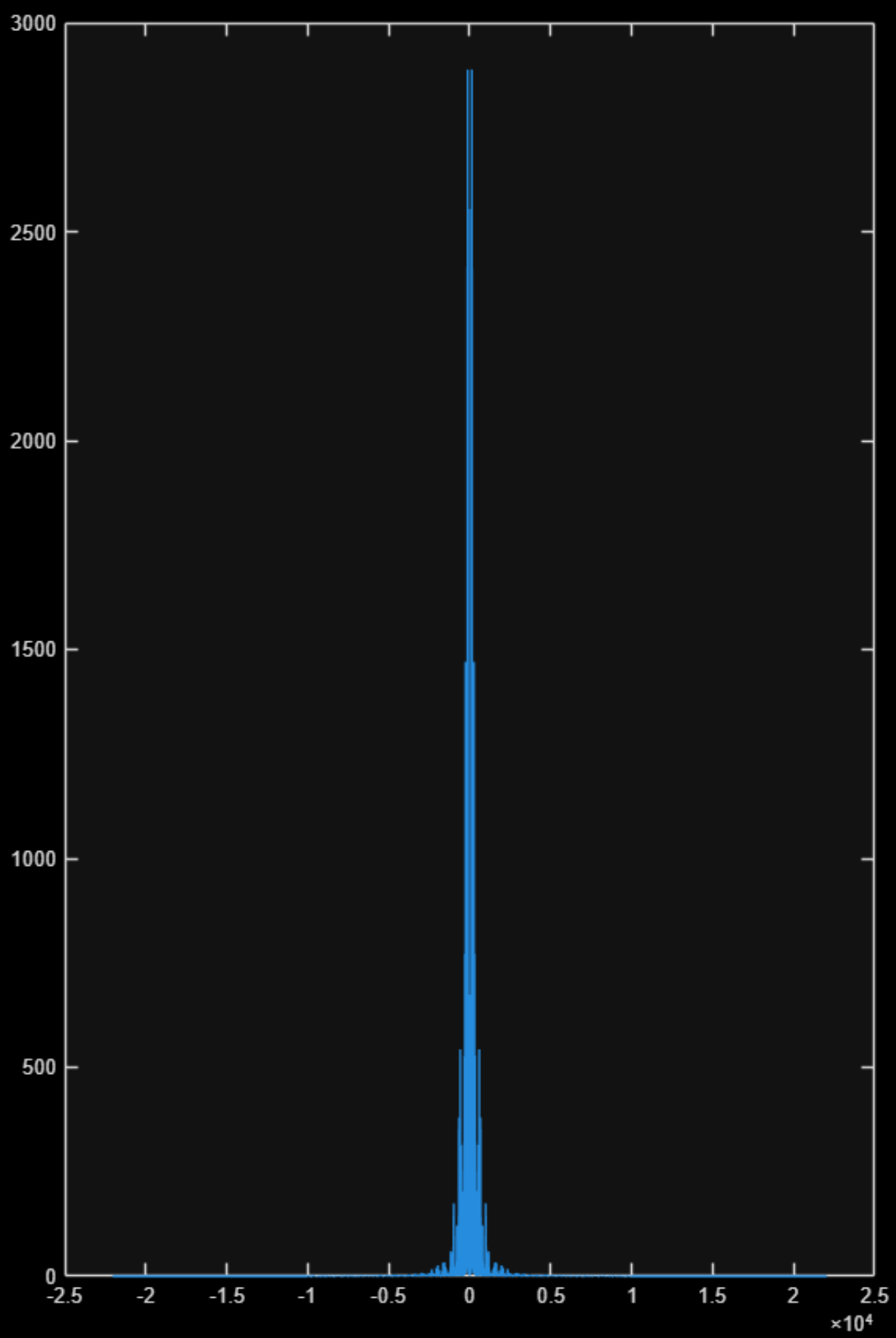
```
OP2 = fir1(99,0.1);  
[h2,w] = freqz(OP2,1,512);  
figure(4); plot(f*w/(2*pi),abs(h2));  
music_fir = filter(OP2,1,music);  
audiowrite('icing2_fir.wav', music_fir, f);
```

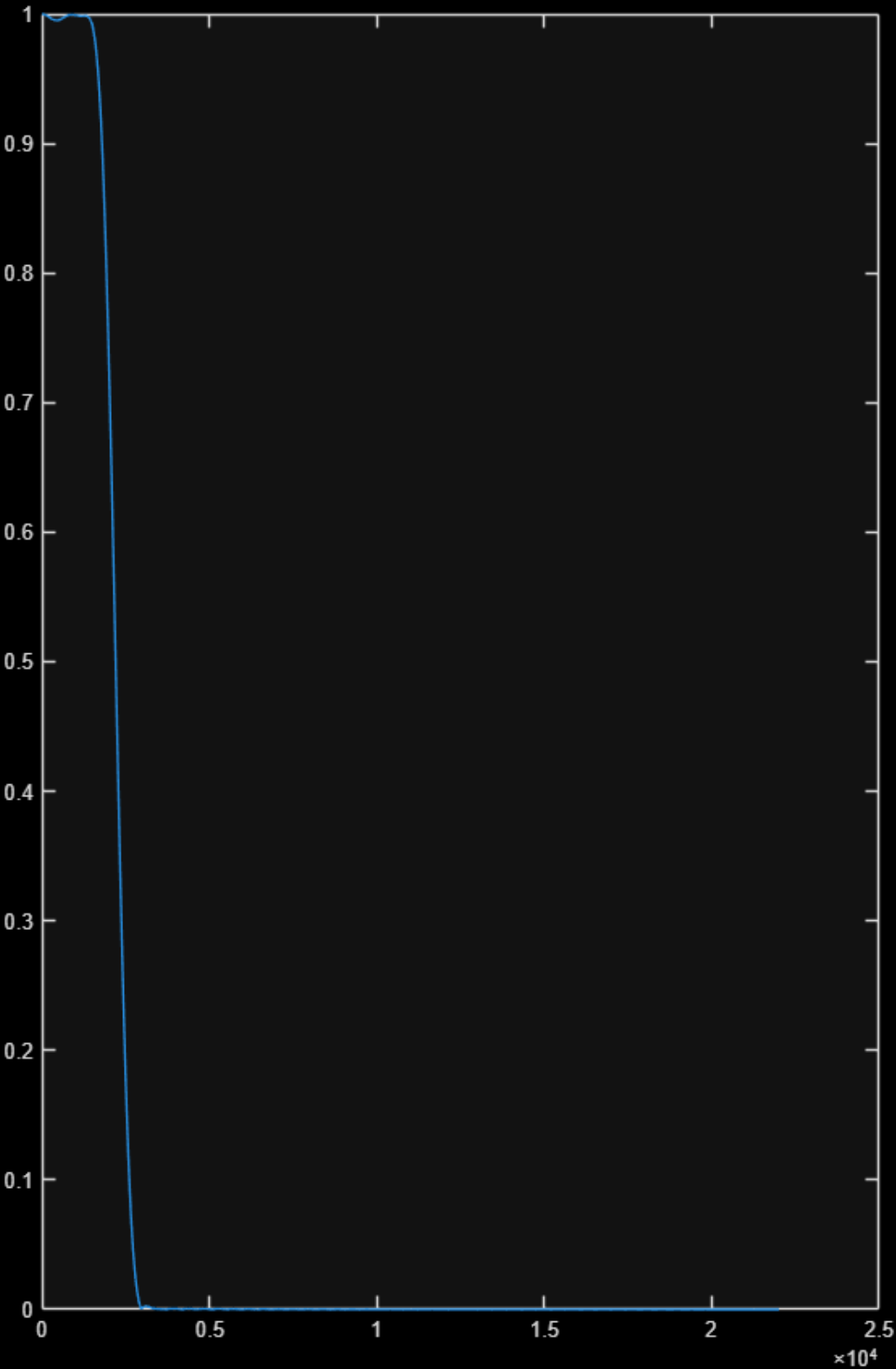
```
fmusic_fir = fft(music_fir);  
fmusic_fir2 = fftshift(fmusic_fir);  
figure(5); plot(hz2,abs(fmusic_fir2));
```

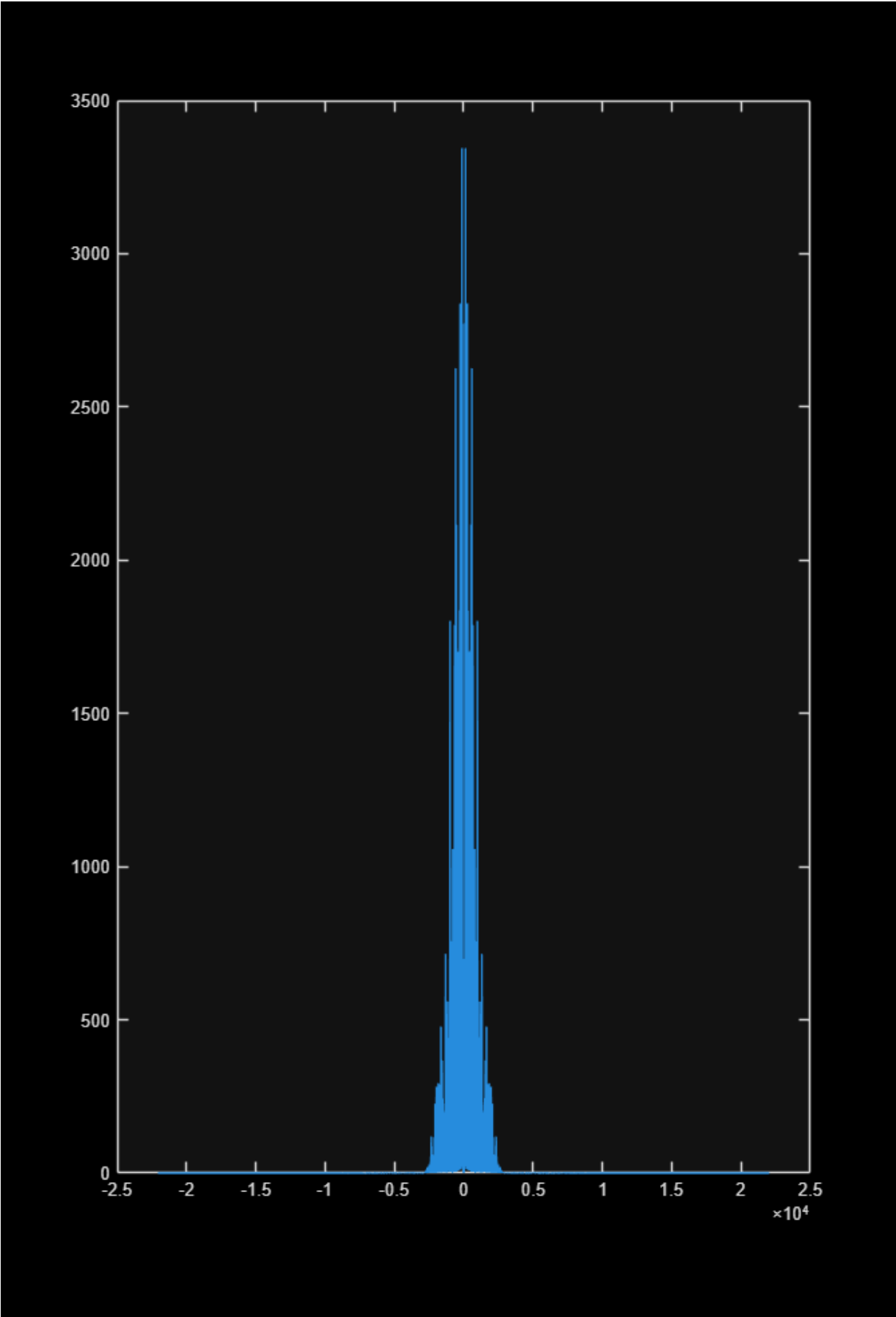
```
% -----
```









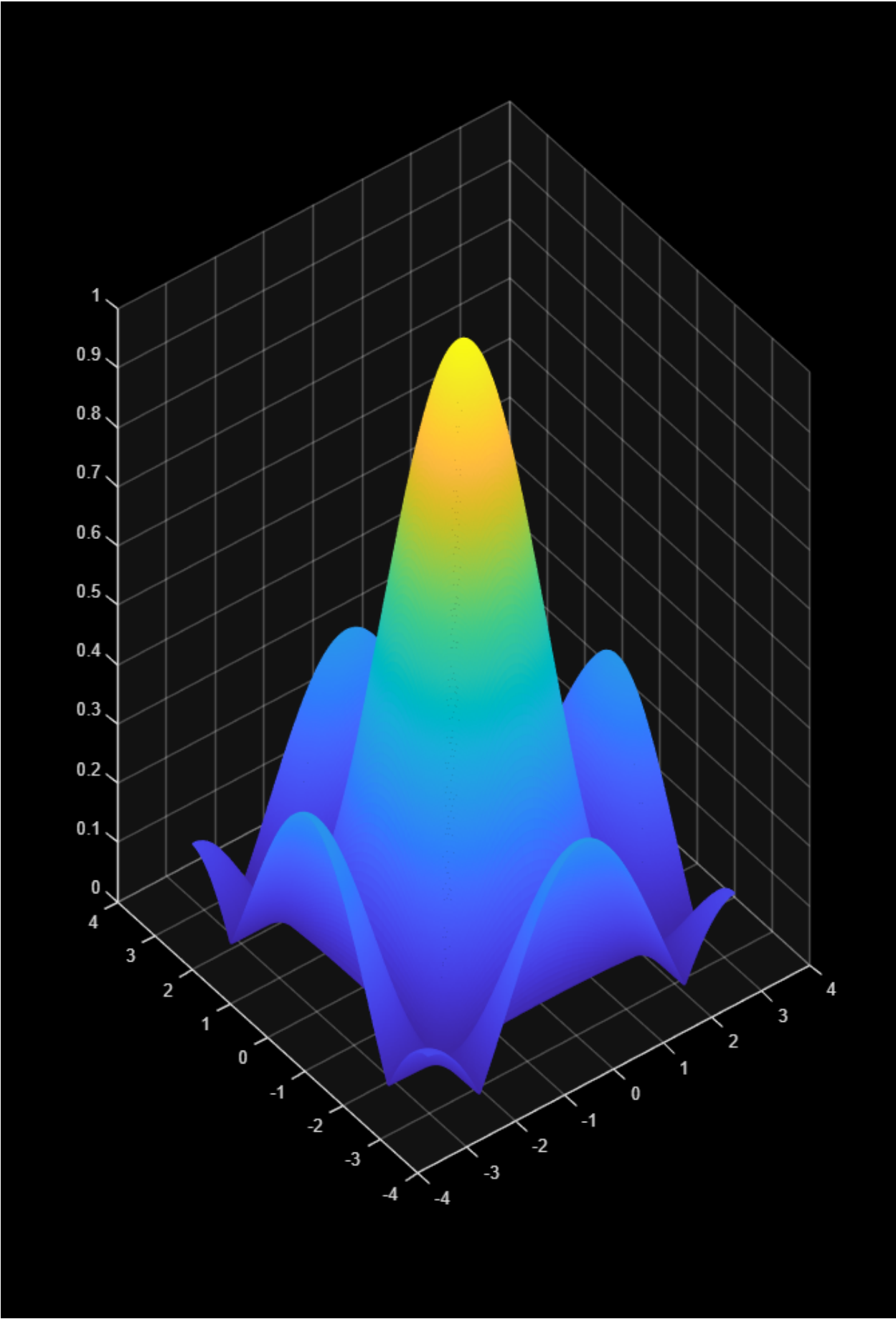


ZADANIE 2

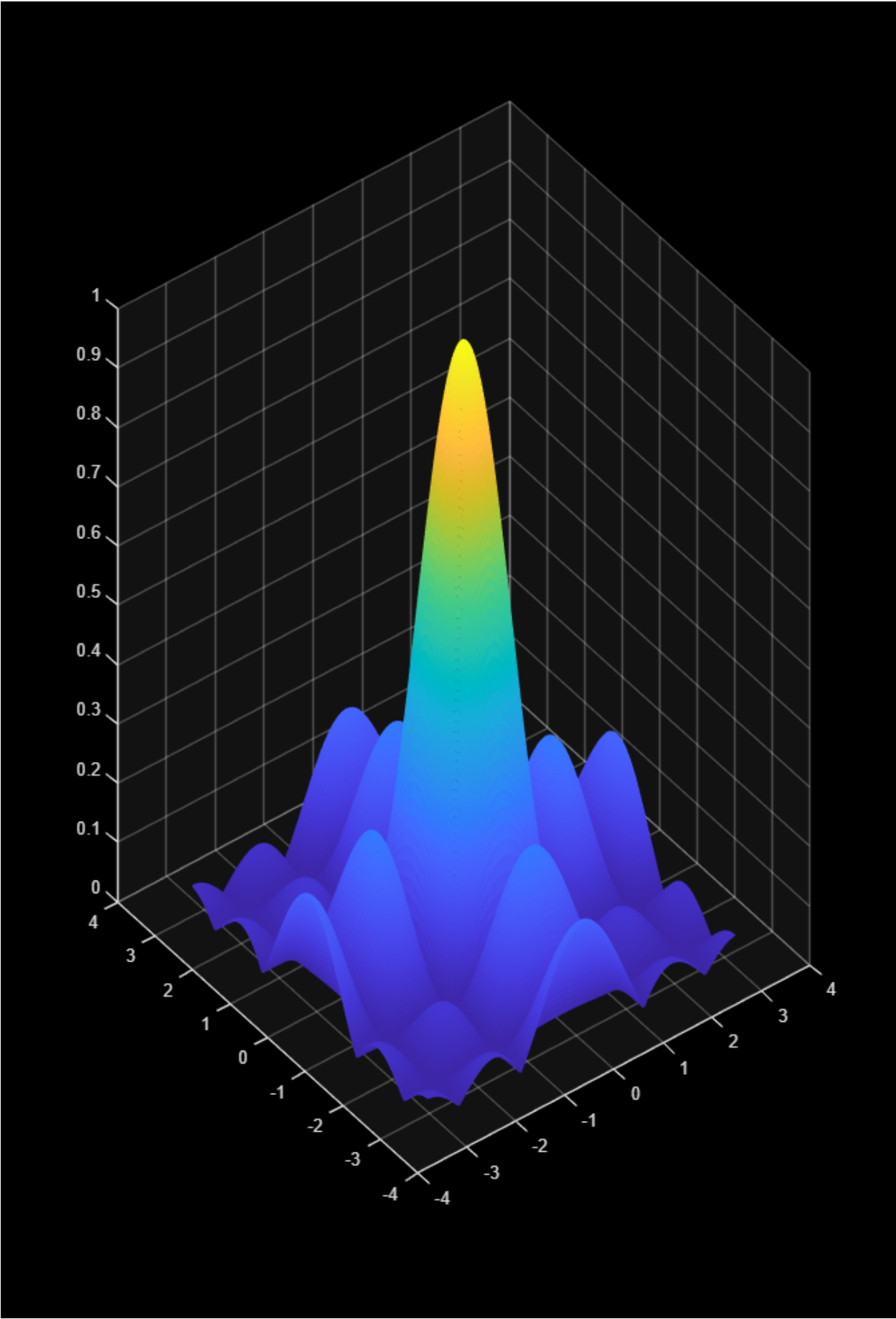
```
zad2( 'LAKE.bmp', 3, 3, 6);  
zad2( 'LAKE.bmp', 5, 5, 10);  
zad2( 'LAKE.bmp', 7, 7, 12);  
zad2( 'LAKE.bmp', 15, 15, 14);  
zad2( 'LAKE.bmp', 3, 5, 16);
```

% -----

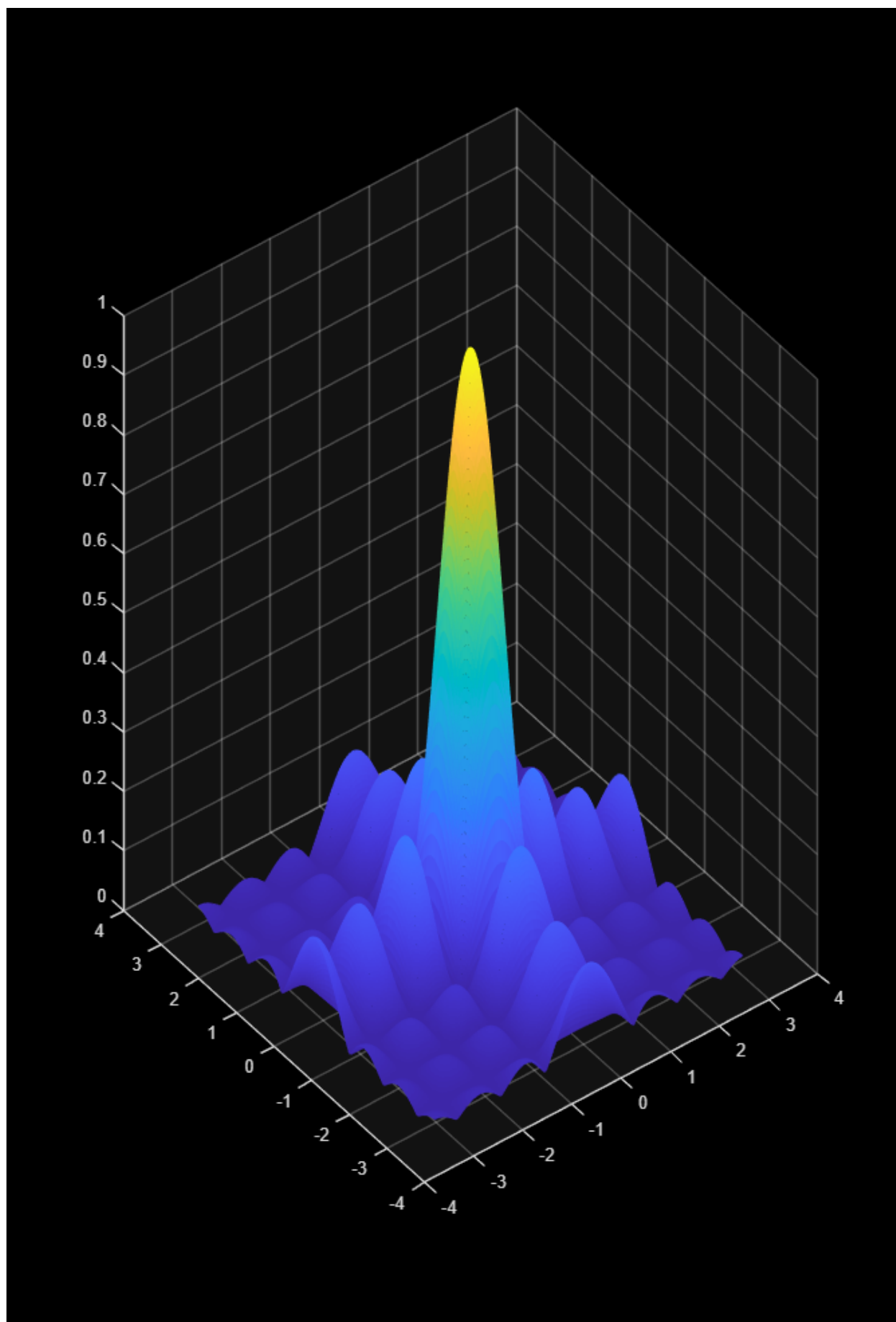




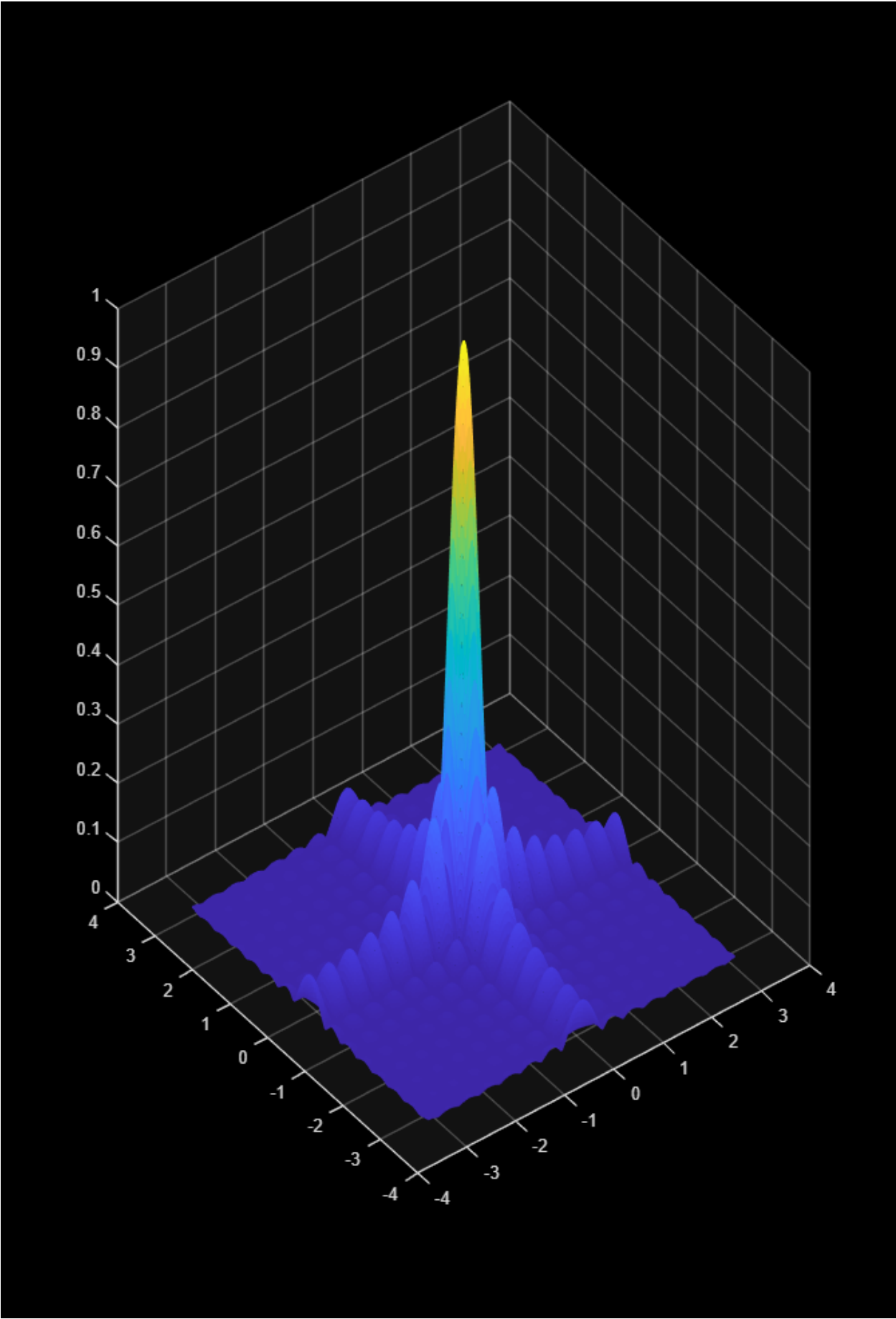




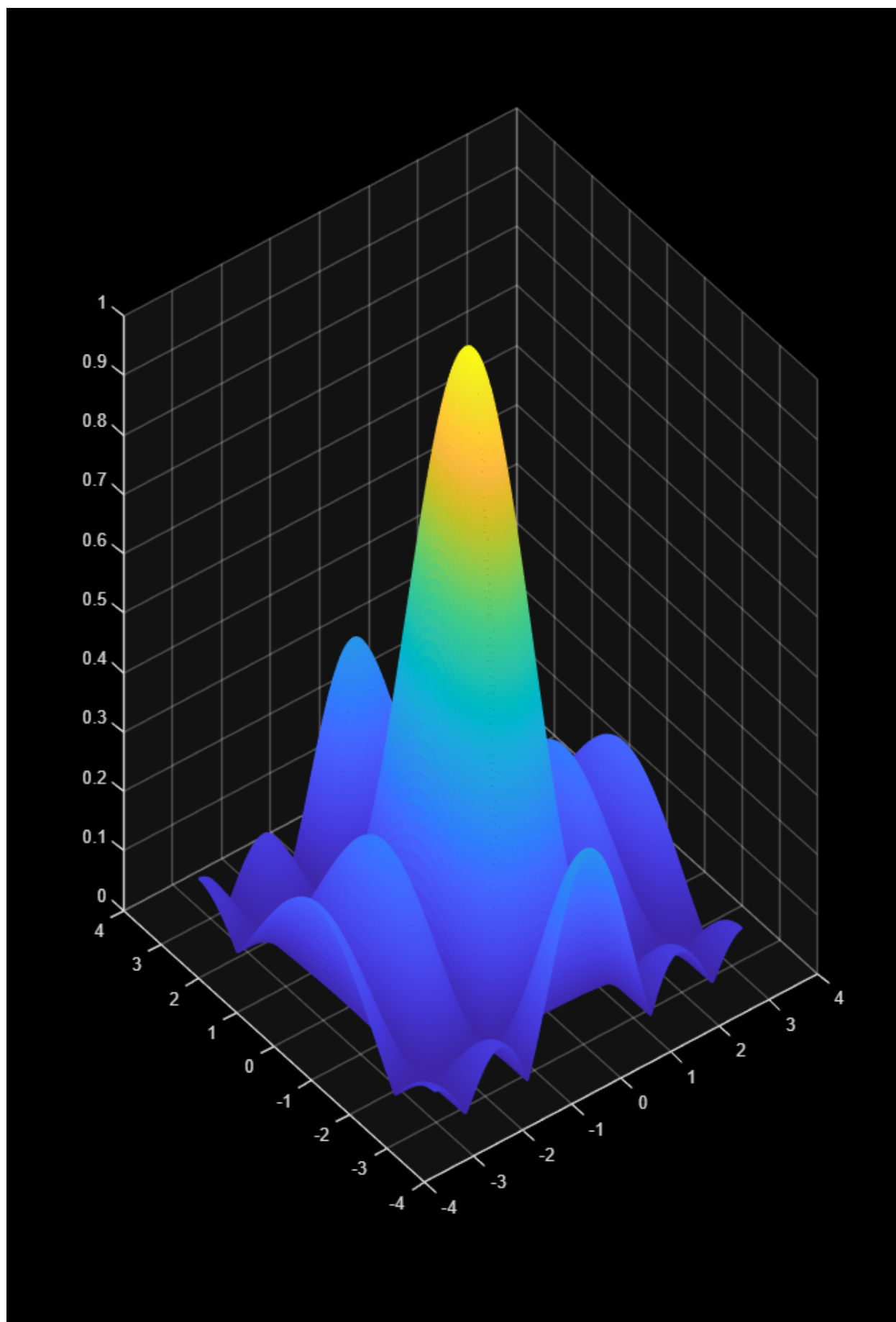












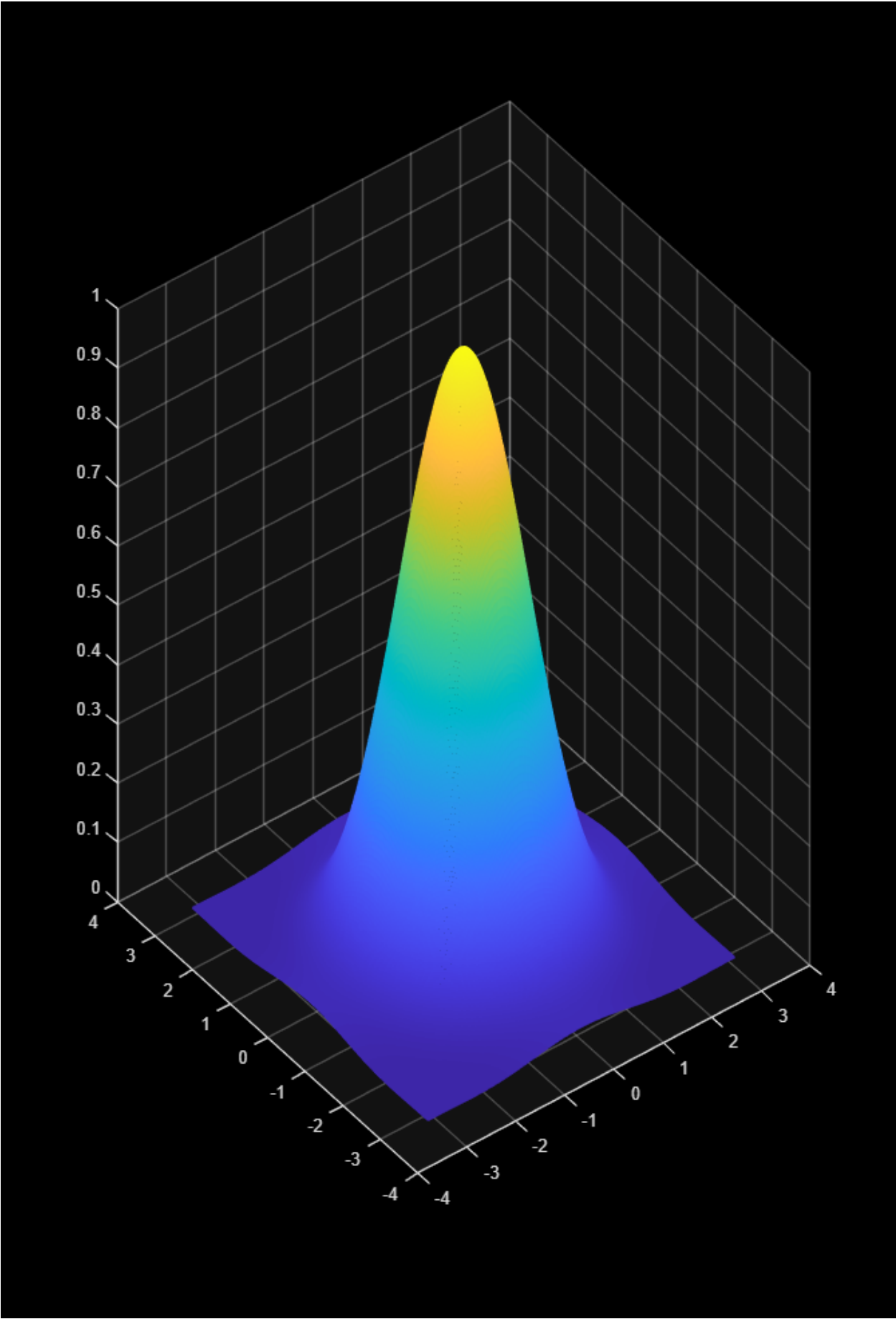
ZADANIE 3

```
zad3( 'LAKE.bmp', 30);
```

```
% -----
```







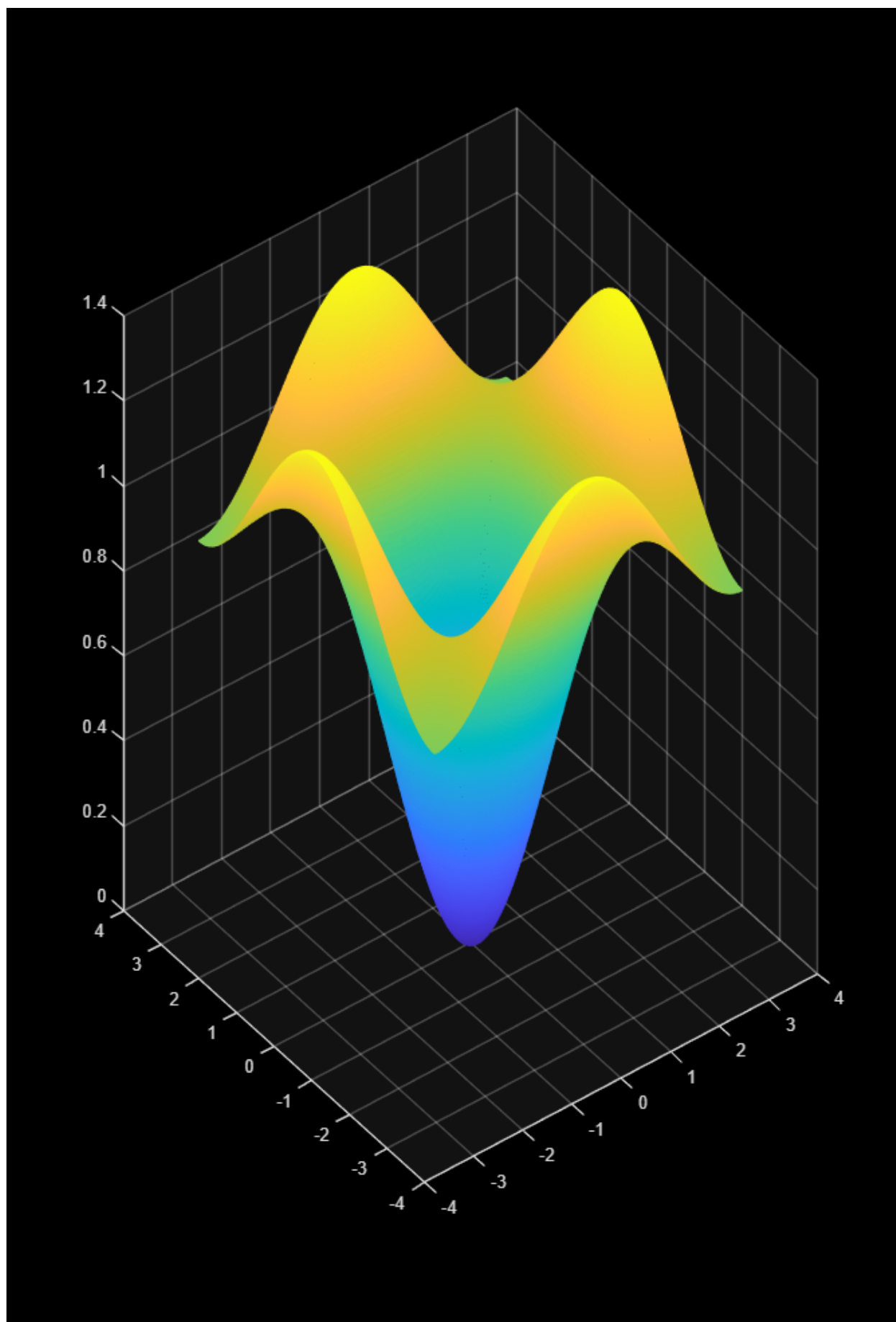
ZADANIE 4

```
zad4( 'LAKE.bmp', 3, 40);  
zad4( 'LAKE.bmp', 5, 44);  
zad4( 'LAKE.bmp', 7, 48);
```

```
% -----
```

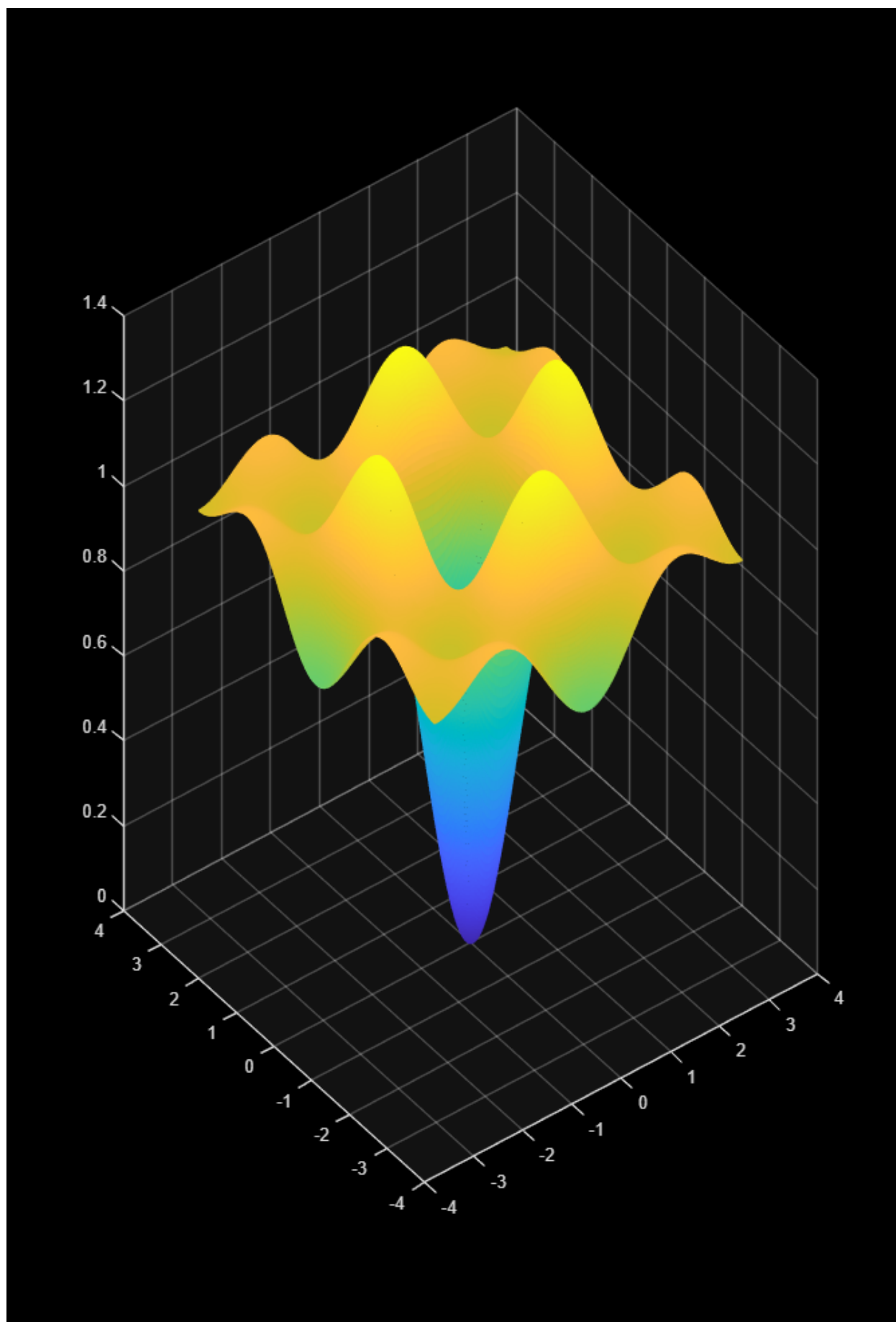






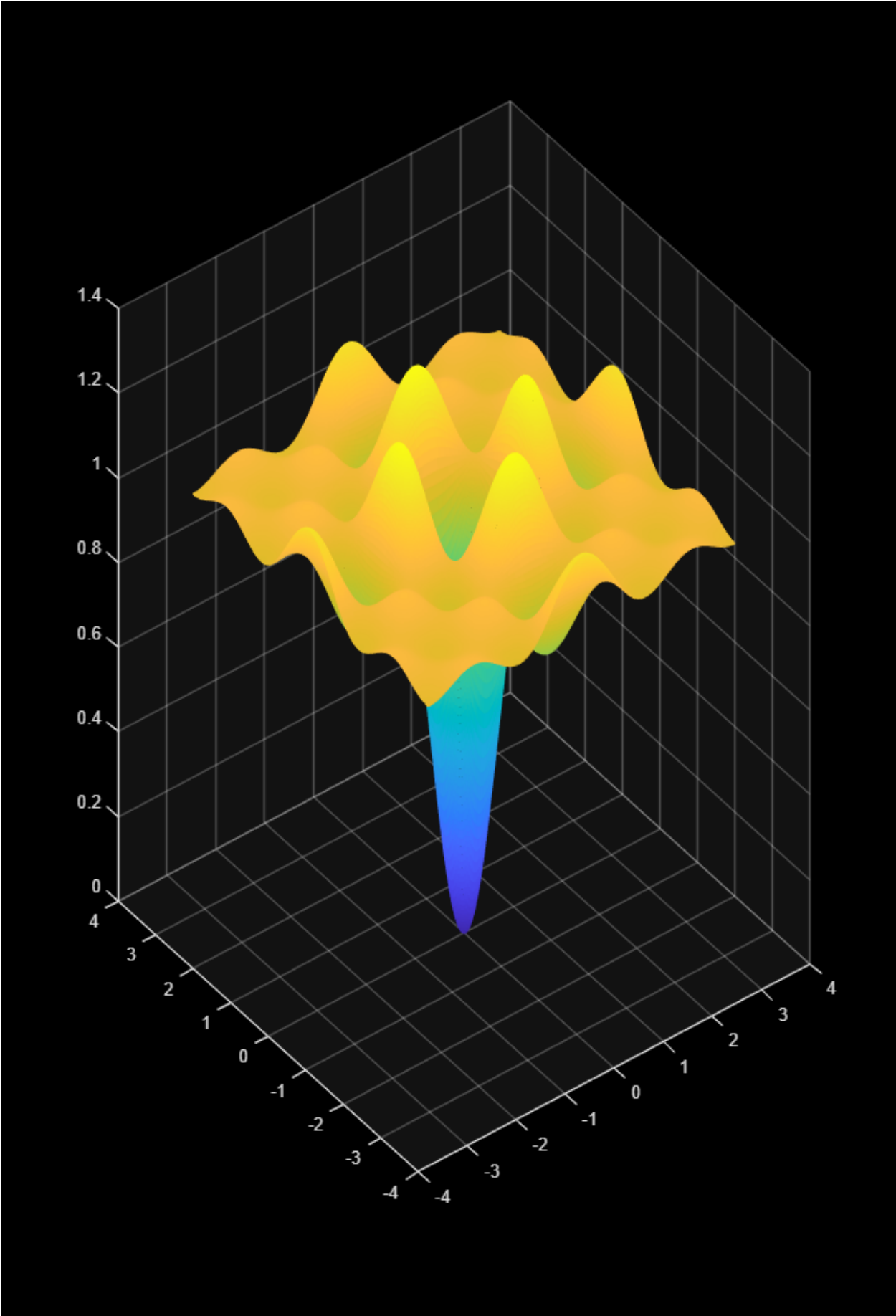






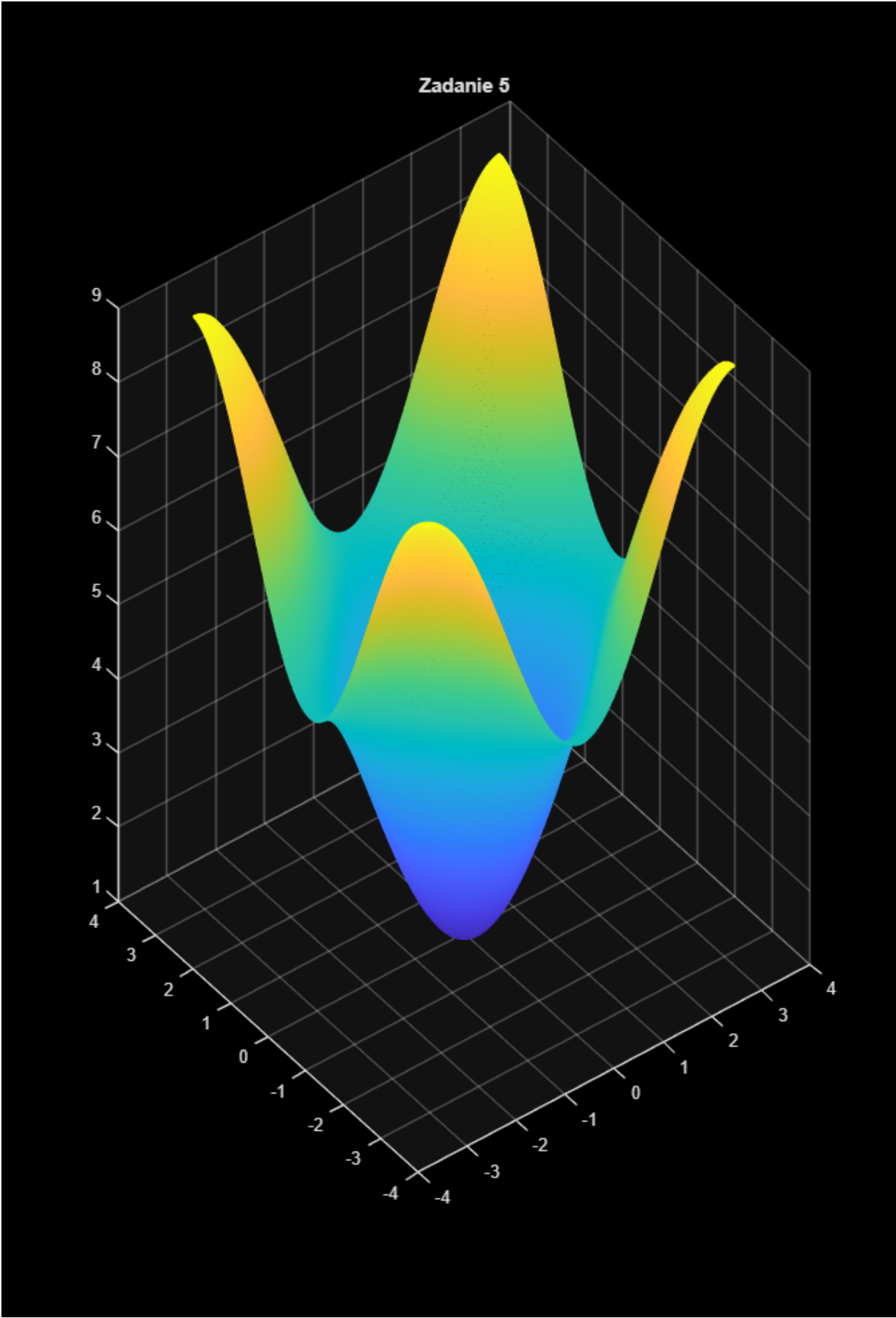


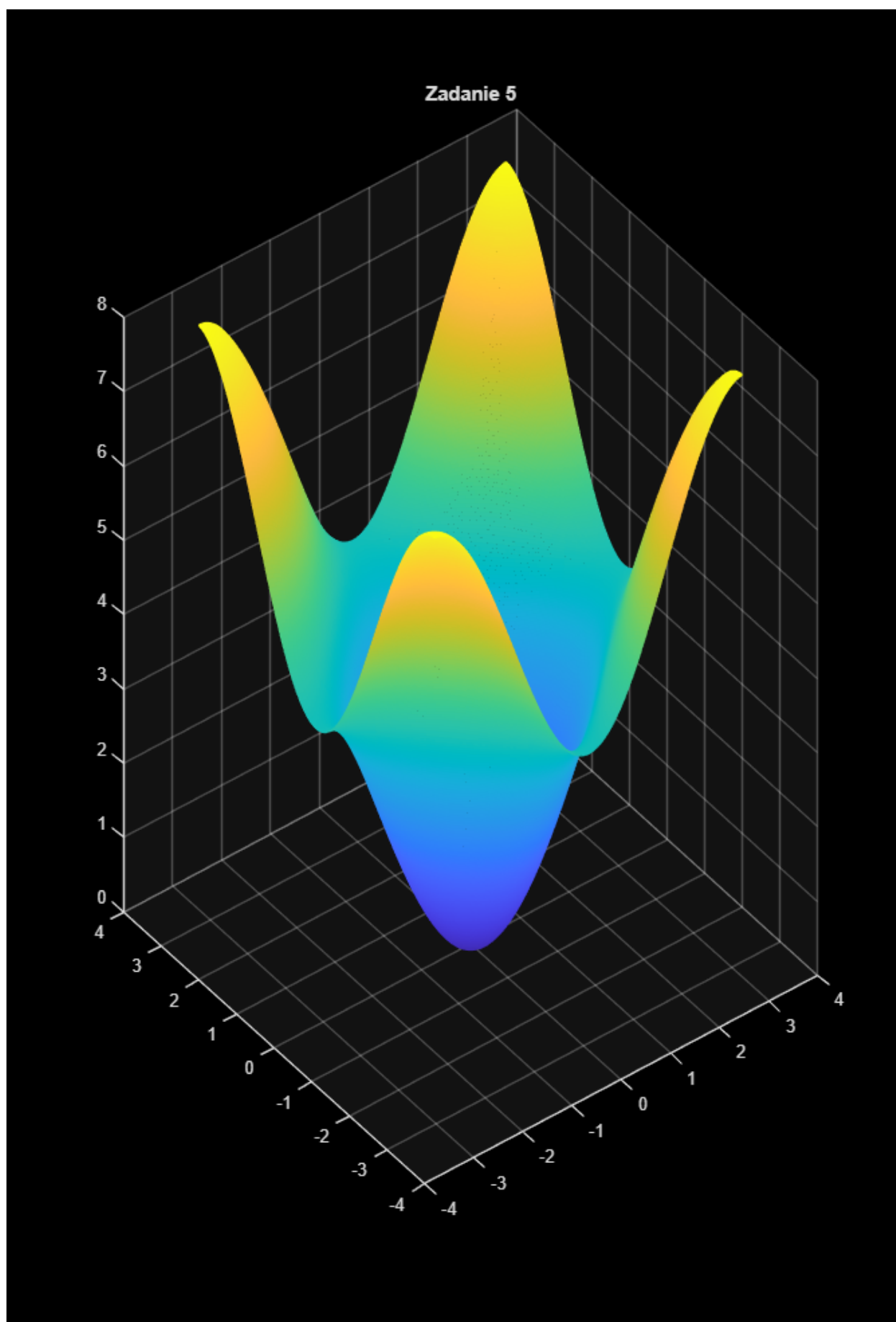




ZADANIE 5

```
-----  
OP1 = [0, -1, 0; -1, 5, -1; 0, -1, 0];  
OP2 = [0, 1, 0; 1, -4, 1; 0, 1, 0];  
[H1, ~, ~] = freqz2(OP1, 256, 256);  
[H2, ~, ~] = freqz2(OP2, 256, 256);  
  
for i = 0:255  
    X(i+1) = i*2*pi/256;  
    Y(i+1) = i*2*pi/256;  
    Xsh(i+1) = i*2*pi/256-pi;  
    Ysh(i+1) = i*2*pi/256-pi;  
end  
figure(111); mesh(Xsh, Ysh, abs(H1));  
title('Zadanie 5');  
figure(222); mesh(Xsh, Ysh, abs(H2));  
title('Zadanie 5');  
% OP1 - filtr wyostrzajacy, wzmacnia krawedzie i szczegoly.  
% OP2 - operator Laplasjanu, filtr gornoprzepustowy do wykrywania krawedzi.  
  
% -----
```





ZADANIE 6

```

zad2('camelsNOISE.bmp',3,3,120); % filtr usredniajacy 3x3
zad2('camelsNOISE.bmp',5,5,122); % filtr usredniajacy 5x5

zad6('camelsNOISE.bmp',3,124); % filtr medianowy 3x3
zad6('camelsNOISE.bmp',5,125); % filtr medianowy 5x5
drawnow;

```

```
% -----
```

FUNKCJE POMOCNICZE

```

function Im1 = zad2(nazwa,M,N,Fig)
% Fig - nr pierwszego obrazka
% MxN - rozmiar operatora
Im = imread(nazwa);
OP = ones(M,N)/(M*N);
[y, x, d] = size(Im);
if d>1 Im = rgb2gray(Im); end
dIm = double(Im);
% Im1 = uint8(zeros(y,x));
Im1 = uint8(filter2(OP,dIm,'same'));
figure(Fig); imshow(Im1);

[H1,fx,fy] = freqz2(OP,[256,256]);
fx = pi*fx; fy = pi*fy;
figure(Fig+1); mesh(fx,fy,abs(H1));
end

function zad3(nazwa,Fig)
% NxN rozmiar operatora
Im = imread(nazwa);
OP = [ 0.003, 0.013, 0.022, 0.013, 0.003;
       0.013, 0.059, 0.097, 0.059, 0.013;
       0.022, 0.097, 0.159, 0.097, 0.022;
       0.013, 0.059, 0.097, 0.059, 0.013;
       0.003, 0.013, 0.022, 0.013, 0.003];
[y, x, d] = size(Im);
if d>1 Im = rgb2gray(Im); end
figure(1); imshow(Im);

dIm = double(Im);
%Im1 = uint8(zeros(y,x));
Im1 = uint8(filter2(OP,dIm,'same'));
figure(Fig); imshow(Im1);

```

```

[H1,~,~] = freqz2(OP,256,256);
for i = 0:255
    X(i+1) = i*2*pi/256;
    Y(i+1) = i*2*pi/256;
    Xsh(i+1) = i*2*pi/256-pi;
    Ysh(i+1) = i*2*pi/256-pi;
end
figure(Fig+1); mesh(Xsh,Ysh,abs(H1));
end

function zad4(nazwa,N,Fig)
% NxN rozmiar operatora
Im = imread(nazwa);
TEMP = ones(N,N)/(N*N);
AUX = zeros(N,N); AUX(ceil(N/2),ceil(N/2)) = 1;
OP = AUX - TEMP;
[y,x,d] = size(Im);
if d>1 Im = rgb2gray(Im); end
figure(Fig); imshow(Im);

dIm = double(Im);
Im1 = uint8(zeros(y,x));
Im1 = uint8(filter2(OP,dIm,'same'));
figure(Fig+1); imshow(Im1);

[H1,~,~] = freqz2(OP,256,256);
for i = 0:255
    X(i+1) = i*2*pi/256;
    Y(i+1) = i*2*pi/256;
    Xsh(i+1) = i*2*pi/256-pi;
    Ysh(i+1) = i*2*pi/256-pi;
end
figure(Fig+2); mesh(Xsh,Ysh,abs(H1));
end

function zad6(nazwa,N,Fig)
% NxN rozmiar operatora medianowego
Im = imread(nazwa);
A = zeros(N,N);
[y,x,d] = size(Im);
if d>1 Im=rgb2gray(Im); end
Im1 = uint8(zeros(y,x));
off = floor(N/2);
for j = 1+off:y-off
    for i = 1+off:x-off
        jj = j-off; ii = i-off;
        A(1:N,1:N) = Im(jj:jj+N-1,ii:ii+N-1);
        a = median(A(:));
        Im1(j,i) = a;
    end
end
end

```

```
    figure(Fig); imshow(Im1);  
    drawnow;  
end
```

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